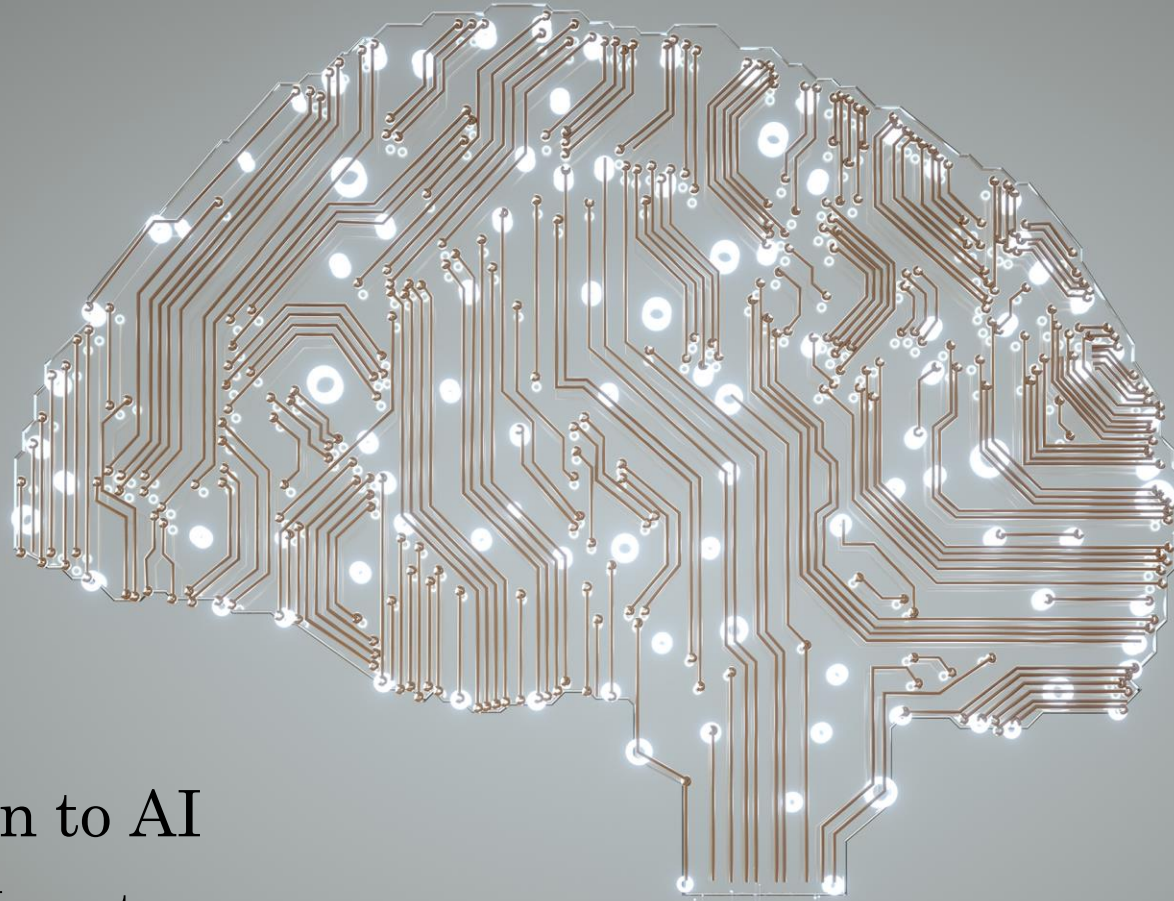
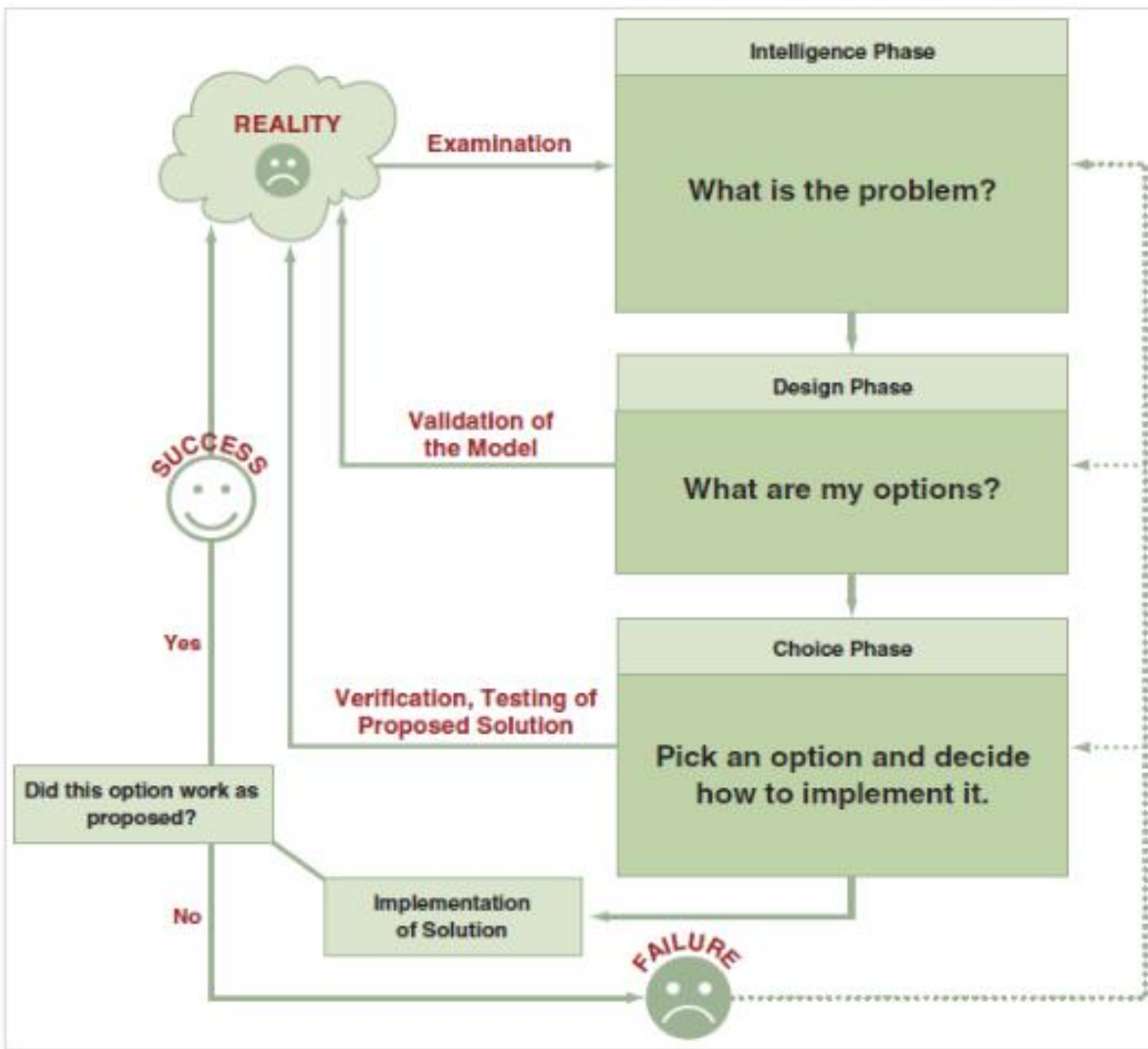


ML & NLP: PGR210



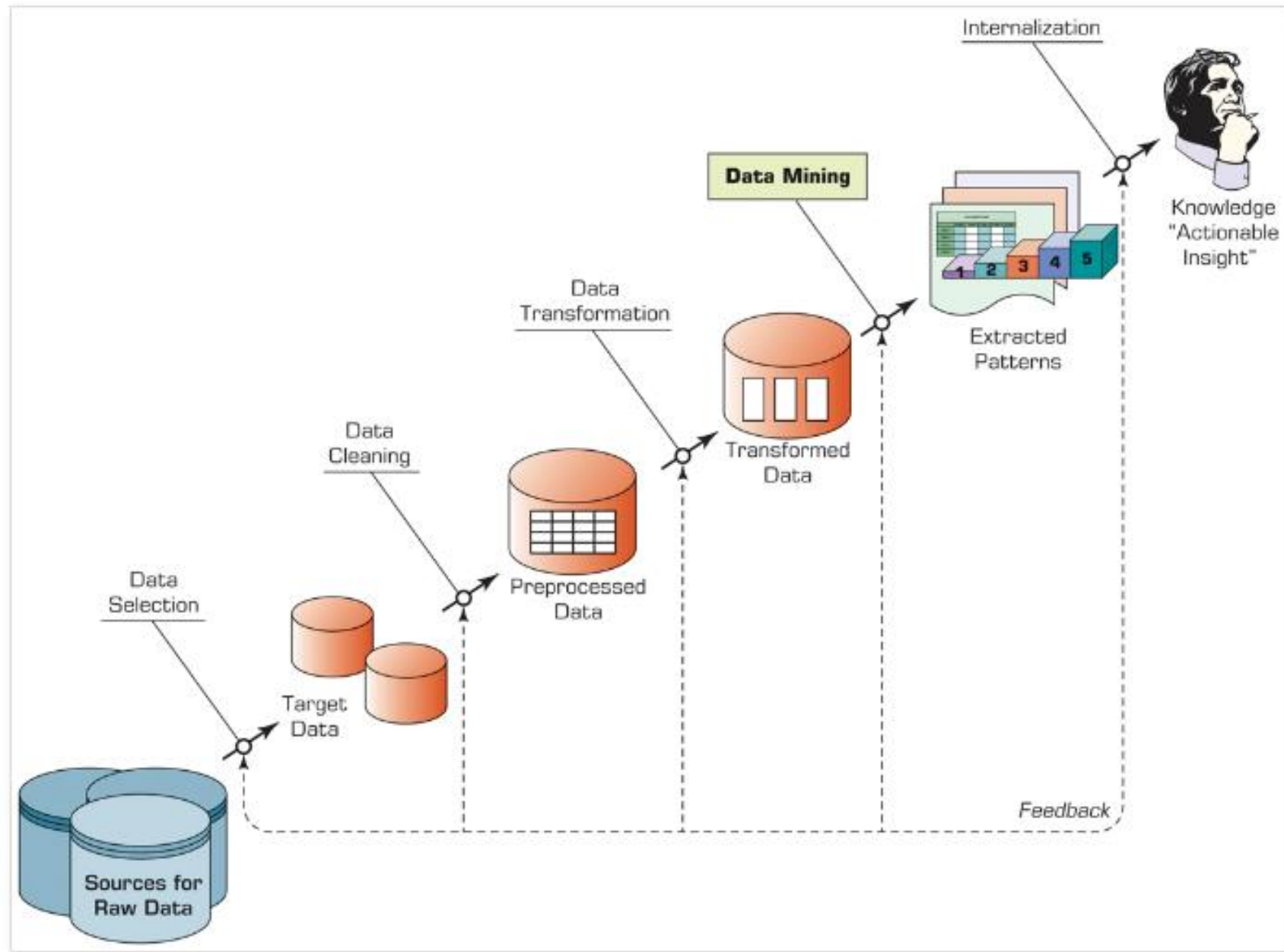
Introduction to AI

Dr. Arvind Keprate

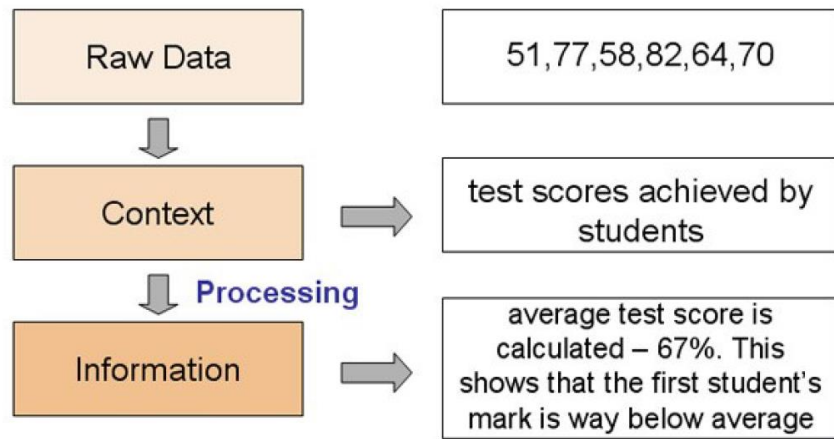


Decision Making

Knowledge Discovery in Databases (KDD) Process



Why Should We Analyze Data?



Data

We started off with the raw data which was 51, 77, 58, 82, 64, 70.

At this stage, those numbers could have been anything from street numbers to chart positions of certain records.

Context

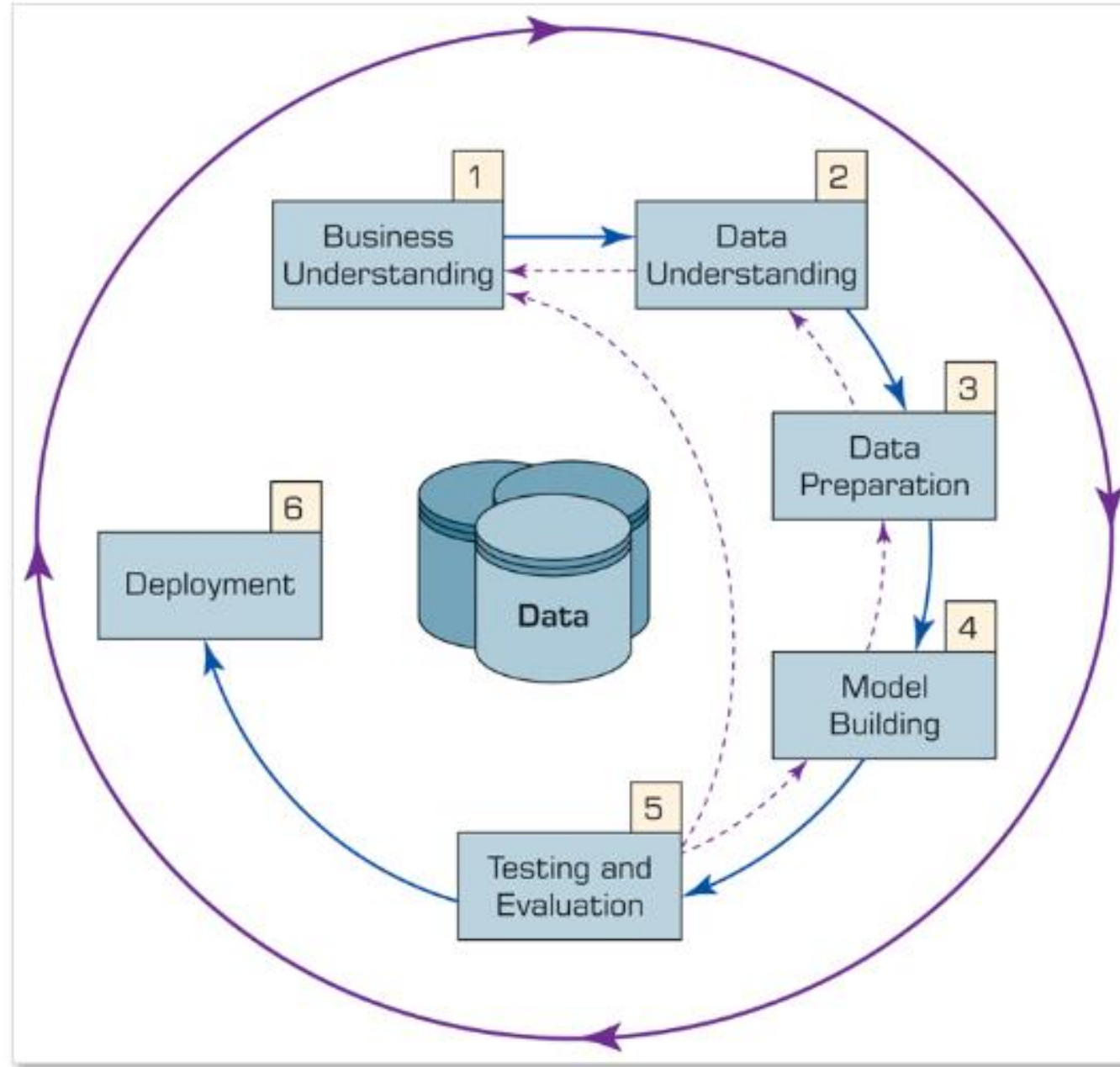
The raw data is given a context. We are told that the numbers are test scores which were achieved by a group of students. At last they start to make some sense.

However even with a context, the data still needs to be processed in order to turn it into information.

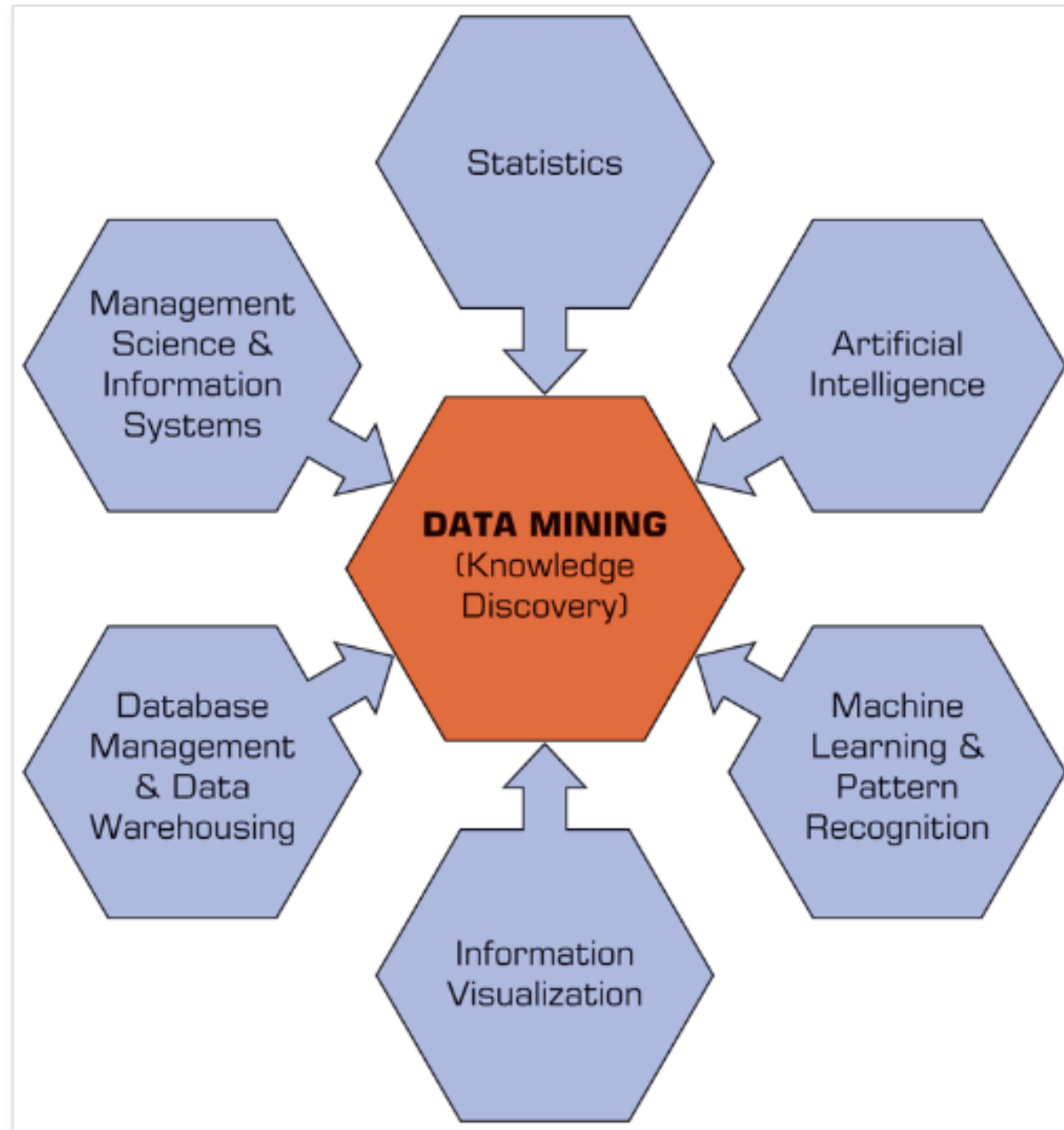
Processing

The processing in this case might be to calculate the average from the set of scores. The average is 67 so we can now tell that student 1 did particularly badly in the test because they were so far below the average mark.

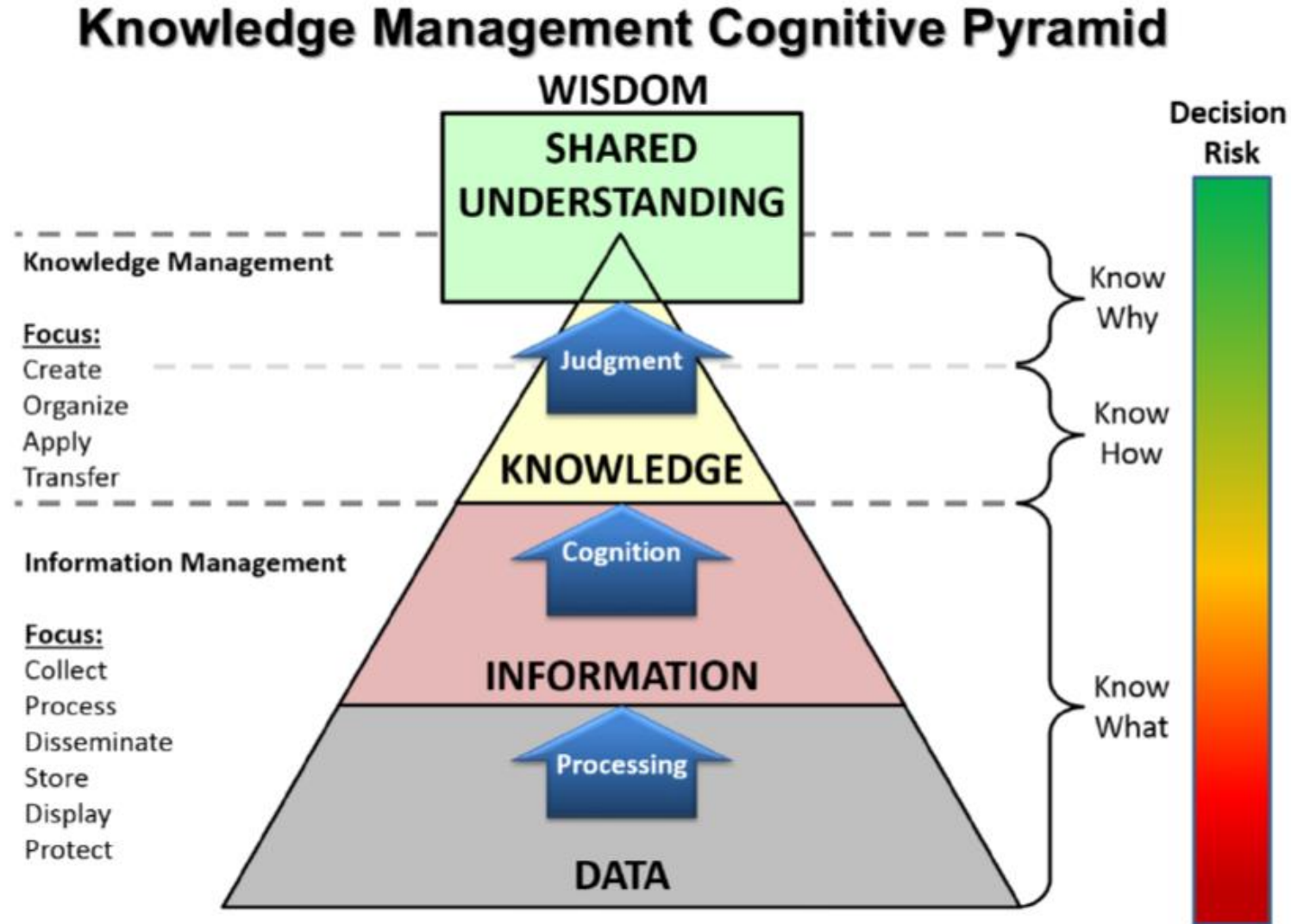
CRISP Data Mining Process



Data Mining



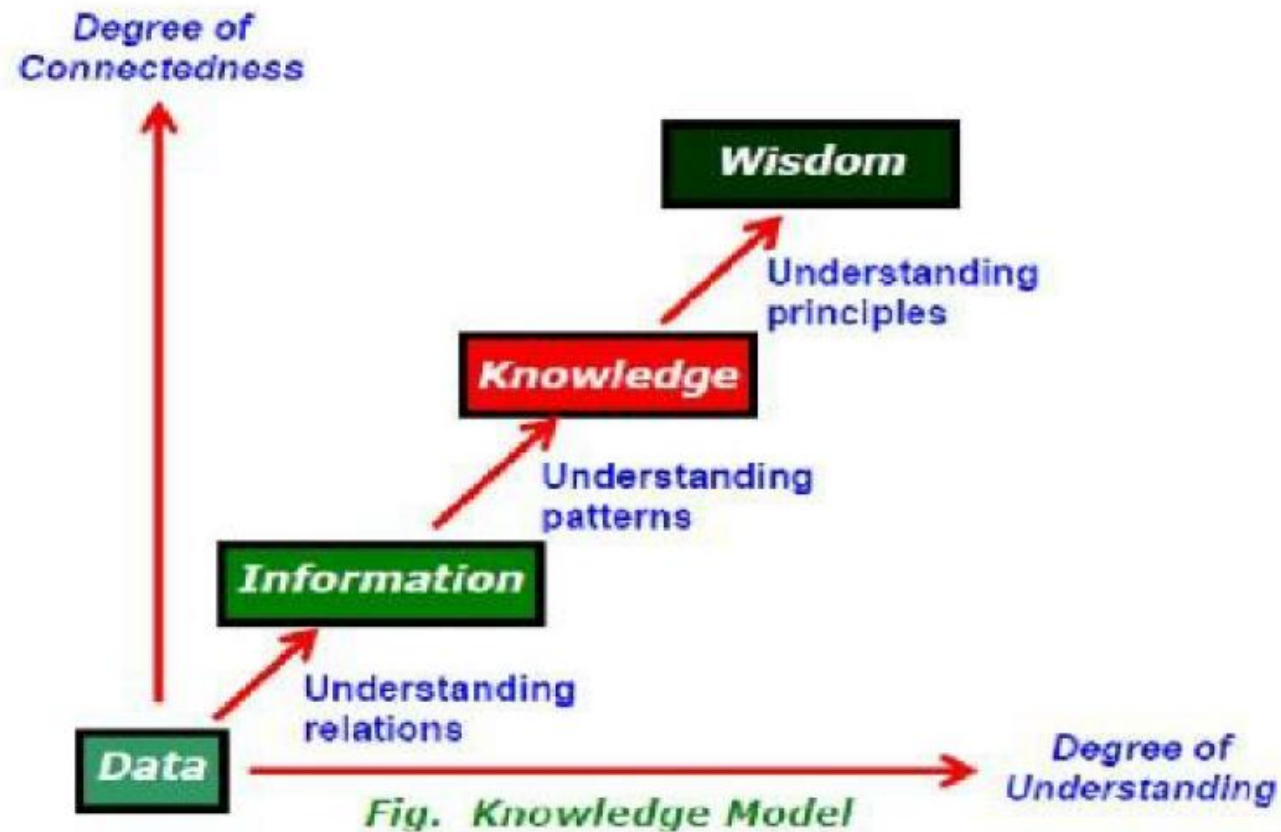
Data Information Knowledge Wisdom (DIKW) Pyramid



DIKW Progression



Fig 1 Knowledge Progression

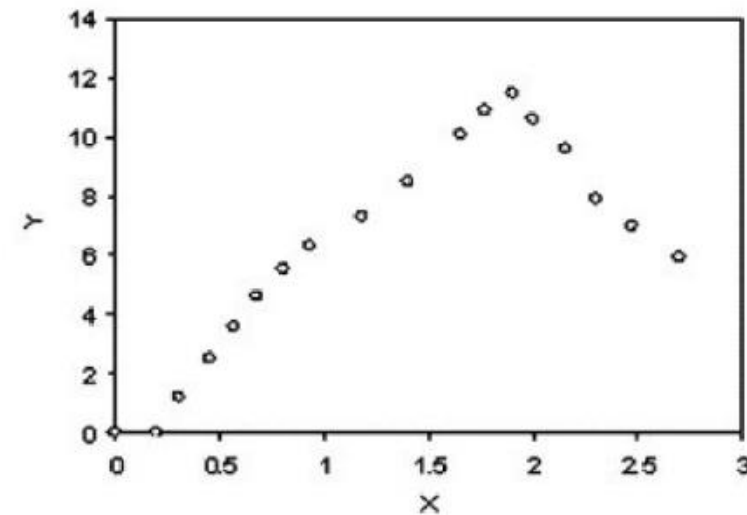


DIKW Example

Experimental Results

Y	X
0	0
0	0.2
1.2	0.3
2.5	0.45
3.6	0.57
4.6	0.68
5.5	0.8
6.3	0.93
7.3	1.18
8.5	1.4
10.1	1.65
10.9	1.77
11.5	1.9
10.6	2
9.6	2.15
7.9	2.3
7	2.47
5.9	2.7

Visualization (Graphs)

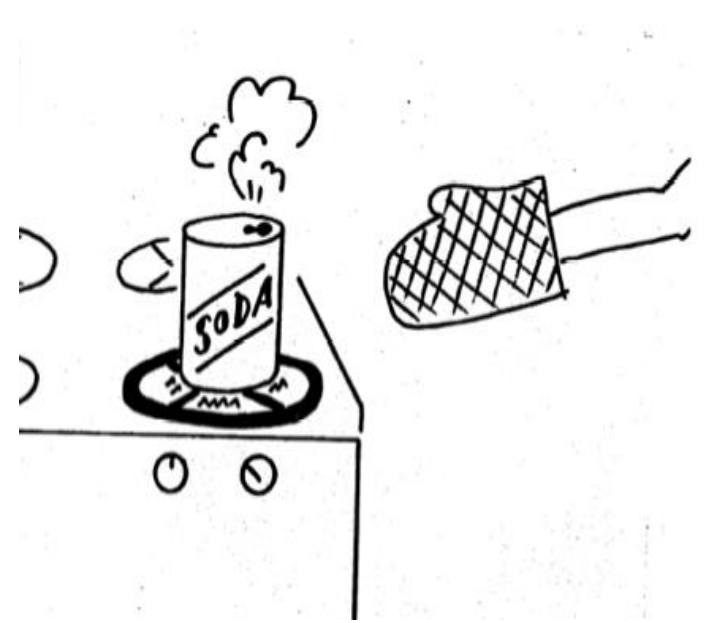


Humanistic
Interpretation

Knowledge (Linguistic Statements)

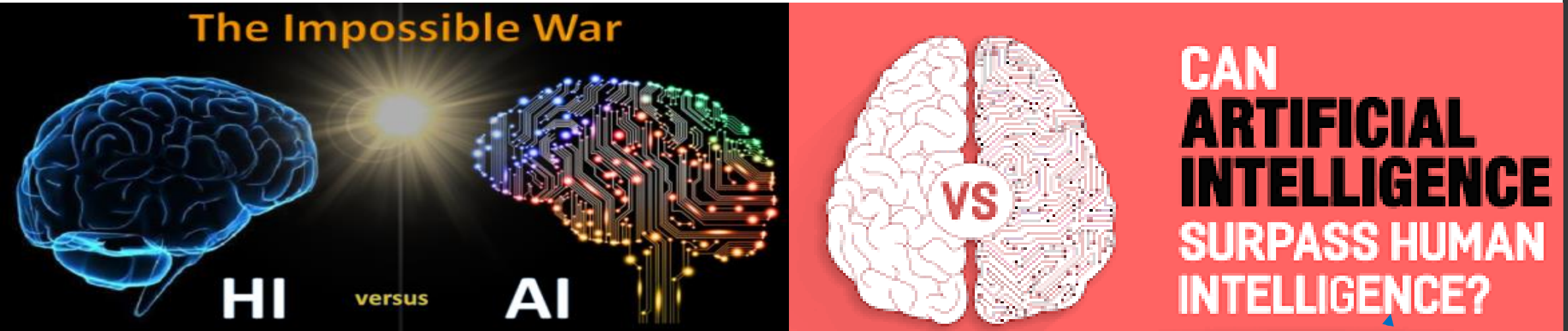
If X is low Then Y is low
If X is moderate Then Y is High
...
Y increases linearly with the increase
in X until X is around 2
...

WHAT IS intelligence



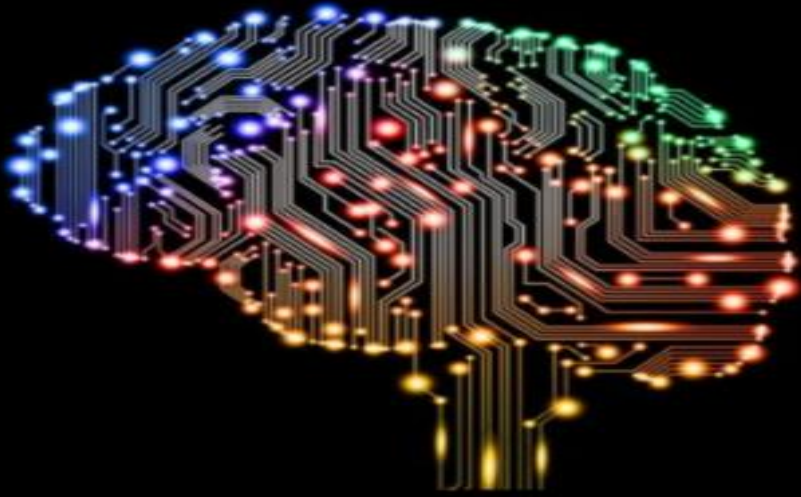
- Remembering your hand was burned the first time is not intelligence.
- Not touching the stove the second time is an example of intelligence, the ability to gain and apply knowledge and skills.
- Intelligence is the ability to take your memory then do something based on the details of the memory [1].

What is Artificial Intelligence (AI)?



- Human intelligence revolves around **adapting** to the **environment** using a **combination** of several **cognitive processes**.
- The field of Artificial Intelligence is a branch of computer science focusing on **designing machines** that can **mimic human behavior** [2].
- AI researchers are able to go as far as implementing **Weak AI**, but not the **Strong AI**. In fact, some believe that Strong AI is never possible due to the various differences between the human brain and a computer.

Artificial Intelligence vs. Human Intelligence



Machine Intelligence

- Fact-based and law-governed: data and algorithms
- Adaptive to environments with abundant data, more dimensions of variables, and greater predictability
- More accurate and effective at deterministic tasks

Human Intelligence

- Experience-based; observation, learning and culture
- Adaptive to environments with high uncertainty and low predictability
- More competent in judgement, creativity

Beginnings

Thresholded
Logic Unit

1943

Perceptron

1957

Adaline

1960

1st Neural Winter

XOR
Problem

1969

AI History Timeline

Multilayer
Backprop

1982

CNNs

1986

LSTMs

1989

1997

2nd Neural Winter

SVMs

1995

GPU Era

Deep
Nets

2006

Alex
Net

2012

1940

1950

1960

1970

1980

1990

2000

2010



S. McCulloch - W. Pitts



R. Rosenblatt



B. Widrow -
M. Hoff



M. Minsky - S. Papert



P. Werbos

D. Rumelhart -
G. Hinton -
R. Williams

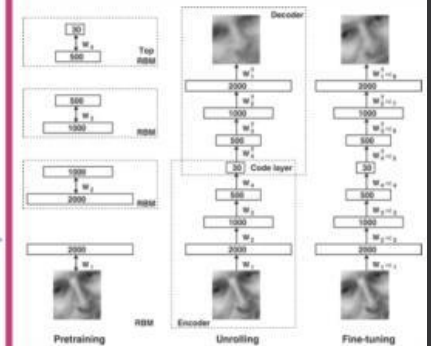
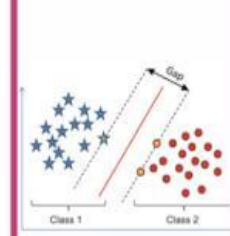
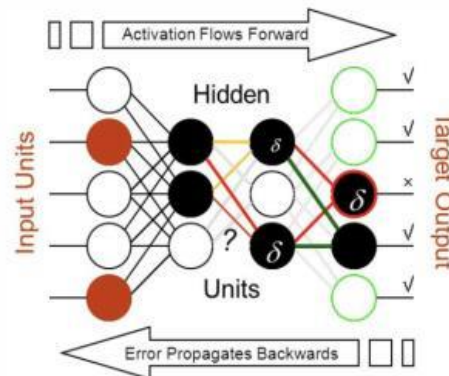
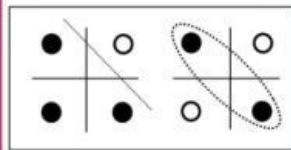
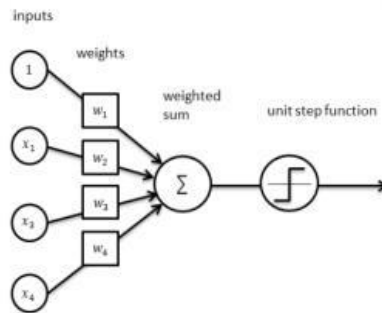
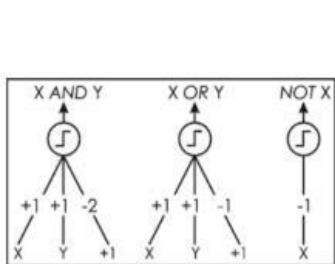
Y. Lecun
J. Schmidhuber



C. Cortes -
V. Vapnik

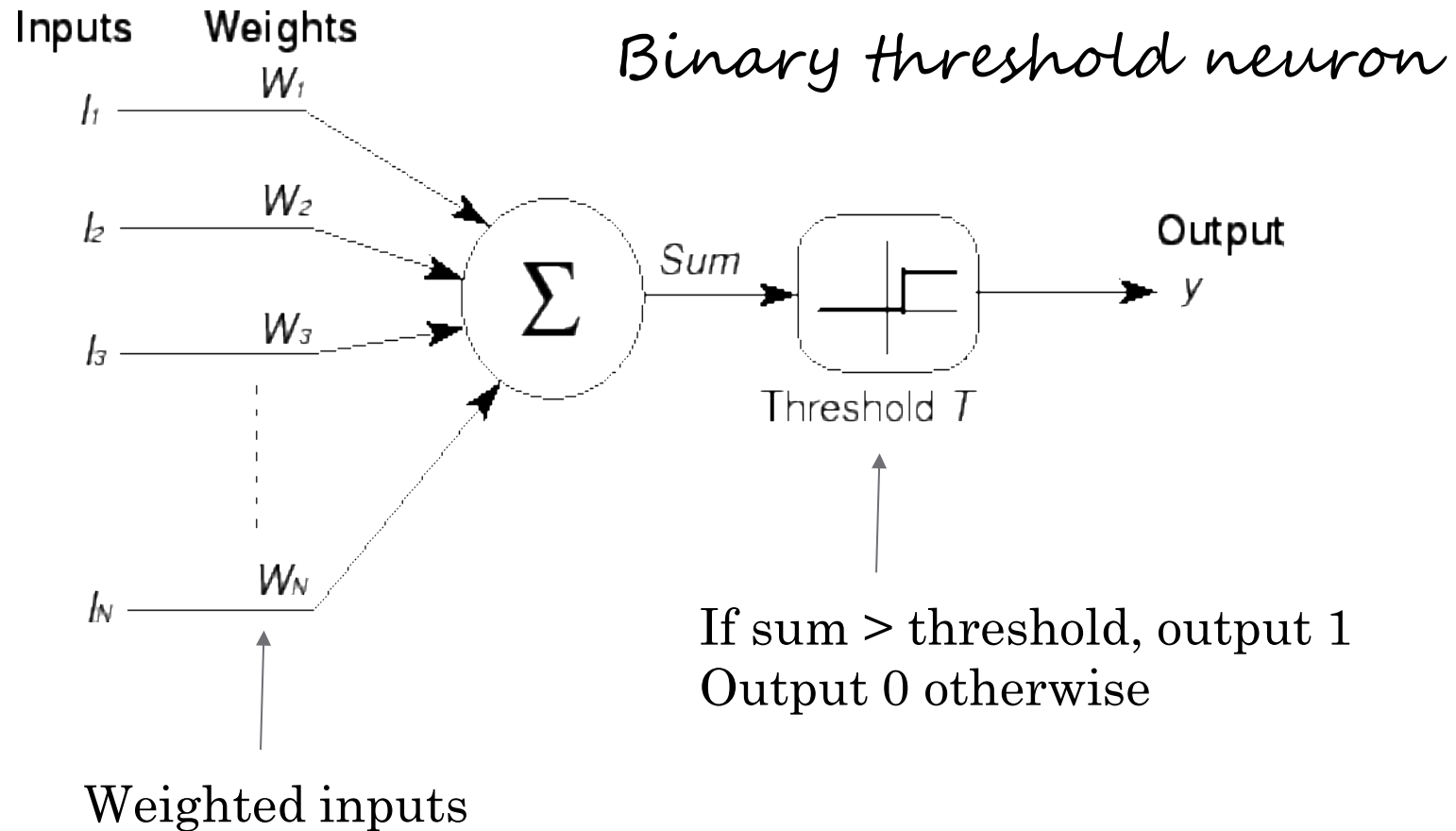


R. Salakhutdinov - J. Hinton -
A. Krizhevsky - I. Sutskever

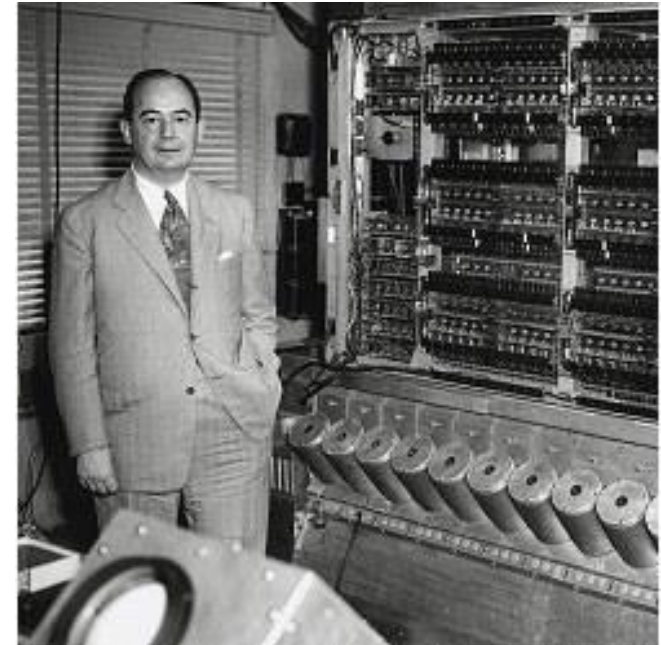


AI History

McCulloch & Pitts - 1943



Von Neumann used this logic
when designing the
“universal computer”



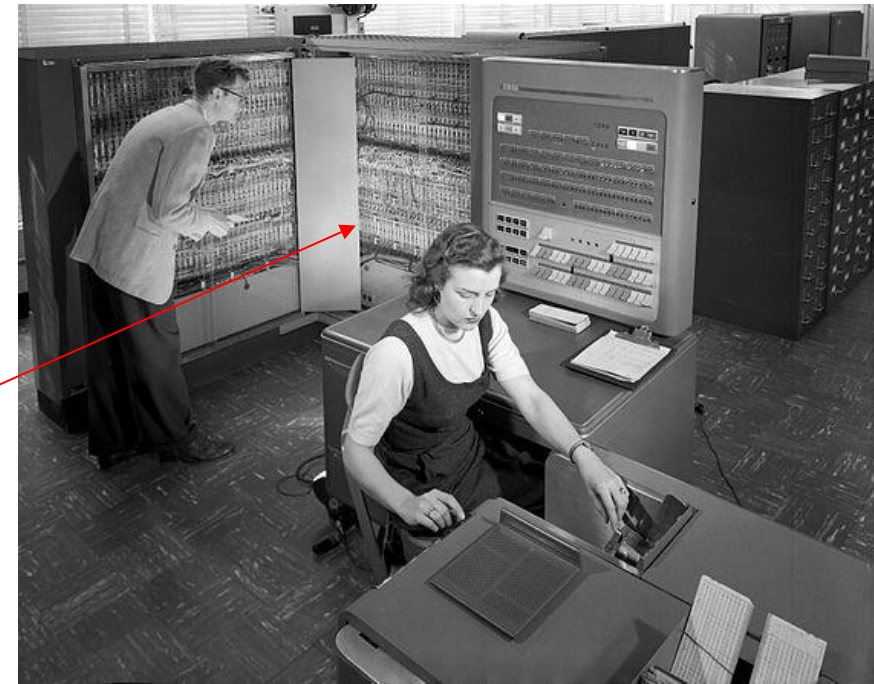
AI History

In the 1950s and 1960s, Principles of Neurodynamics were examined and Symbolic ML expanded



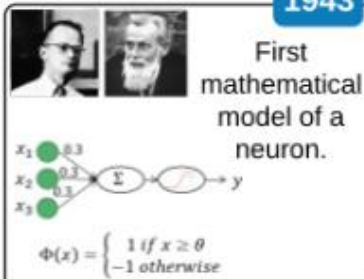
When I show these shapes to the camera

This IBM 704 computer can say "it's a triangle"



AI Evolution

1943



Electronic Brain by McCulloch & Pitts

1950



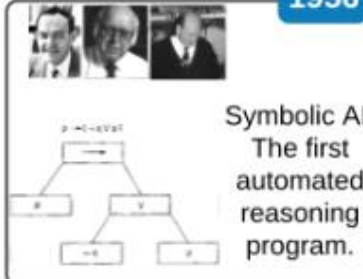
Turing Test by Alan Turing

1952



Checkers Program by Arthur Samuel

1956



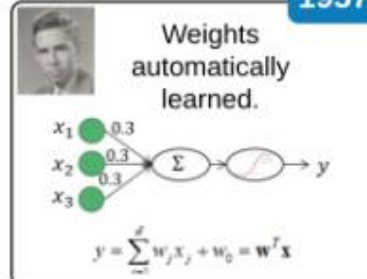
Logic Theorist by Newell, Simon, Shaw

1956



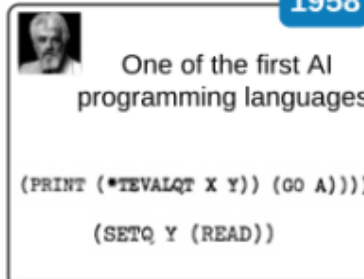
Dartmouth Summer Research Project organized by John McCarthy

1957



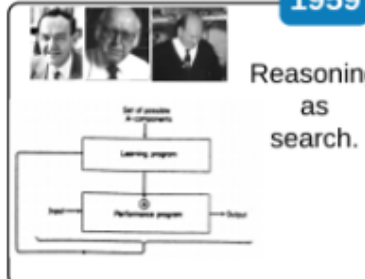
Perceptron by Frank Rosenblatt

1958



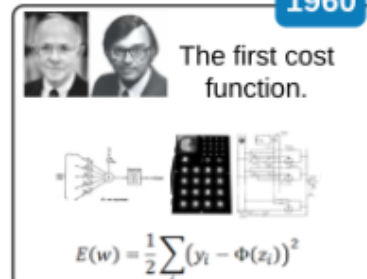
LISP by John McCarthy

1959



General Problem Solver by Newell, Simon, Shaw

1960



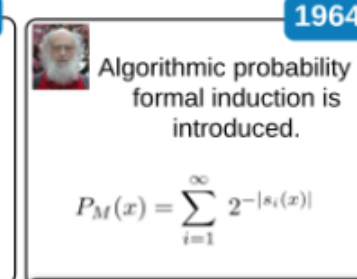
ADALINE by Widrow & Hoff

1964



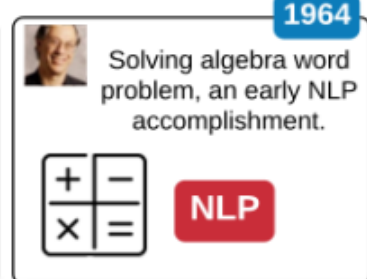
ELIZA by Joseph Weizenbaum

1964



Universal Bayesian Methods by Ray Solomonoff

1964



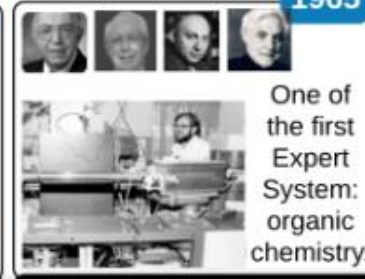
STUDENT by Daniel G. Bobrow

1965



Fuzzy Logic by Lotfi Zadeh

1965



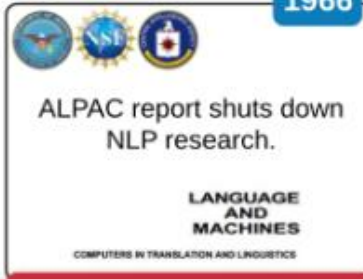
DENDRAL by Feigenbaum, Buchanan, Lederberg, Djerassi

1966



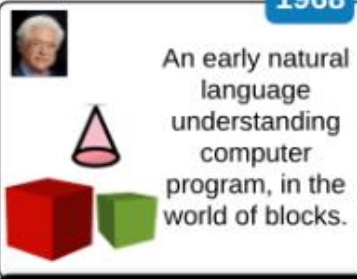
Shakey the Robot by SRI International

1966



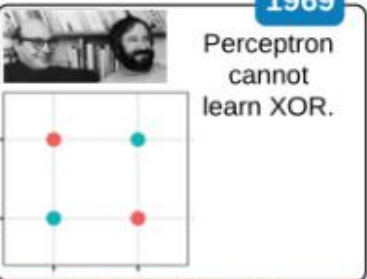
ALPAC Report by ALPAC

1968



SHRDLU by Terry Winograd

1969



XOR Problem by Minsky & Papert

AI Evolution

1970



One of the first medical expert systems.

Please Enter Findings of PAIN ABDOMEN
*GO
ABDOMEN PAIN GENERALIZED ?
NO
ABDOMEN PAIN EPIGASTRIUM ?
NO

INTERNIST-I by Myers, Miller, Pople

1970



Backpropagation & automatic differentiation.

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial w_1} \frac{\partial w_1}{\partial x}$$

Automatic differentiation by Seppo Linnainmaa

1972



One of the first logic programming languages.

```
animal(X) :- cat(X).
```

PROLOG by Colmerauer & Kowalski

1974



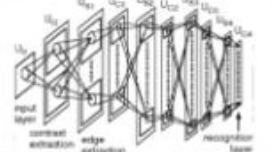
An early backward chaining expert system for medical diagnosis.

MYCIN by Shortliffe, Buchanan, Cohen

1979



The first convolutional neural network (CNN).

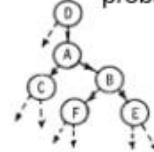


Neocognitron by Kunihiko Fukushima

1982



Foundation of graphical probabilistic models.

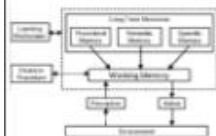


Bayesian Networks by Judea Pearl

1983



A cognitive architecture for general intelligence.

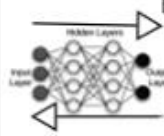


SOAR by Laird, Newell, Rosenbloom

1986



Backpropagation is popularized.

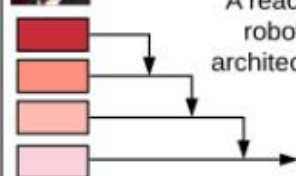


Backpropagation in MLP by Rumelhart, Hinton, Williams

1987



A reactive robotic architecture.

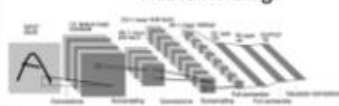


Subsumption by Rodney Brooks

1989



Convolutional neural networks (CNN) used for recognizing handwriting.

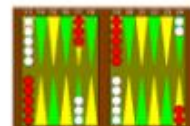


LeNet by Yann LeCun

1992



Almost champion-level backgammon, using reinforcement learning.

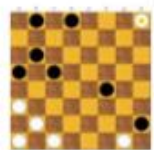


TD-Gammon by Gerald Tesauro

1994



Chinook, draughts player: the first program to win the world champion title against humans.



Chinook by a Team led by Jonathan Schaeffer

1995



Soft-margin SVM is introduced.



Support Vector Machines by S. Vapnik & Cortes

1995



MNIST is born.



MNIST by NIST

1996



DeepBlue beats Kasparov in chess.

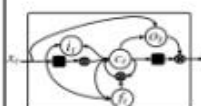


DeepBlue by IBM

1997



LSTM for addressing vanishing gradients.



Long Short-Term Memory (LSTM) by Hochreiter & Schmidhuber

2006



Modern deep learning is born.



Deep Boltzmann Machine by Salakhutdinov & Hinton

2009

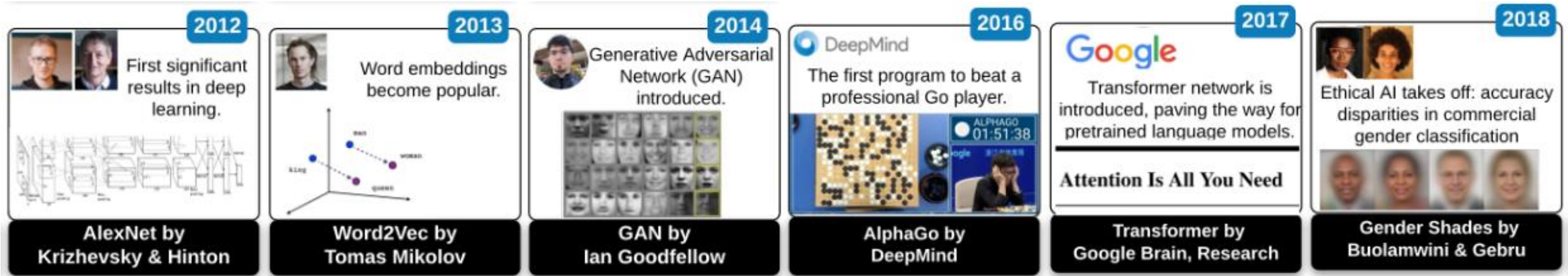


ImageNet, a large-scale image dataset is introduced.

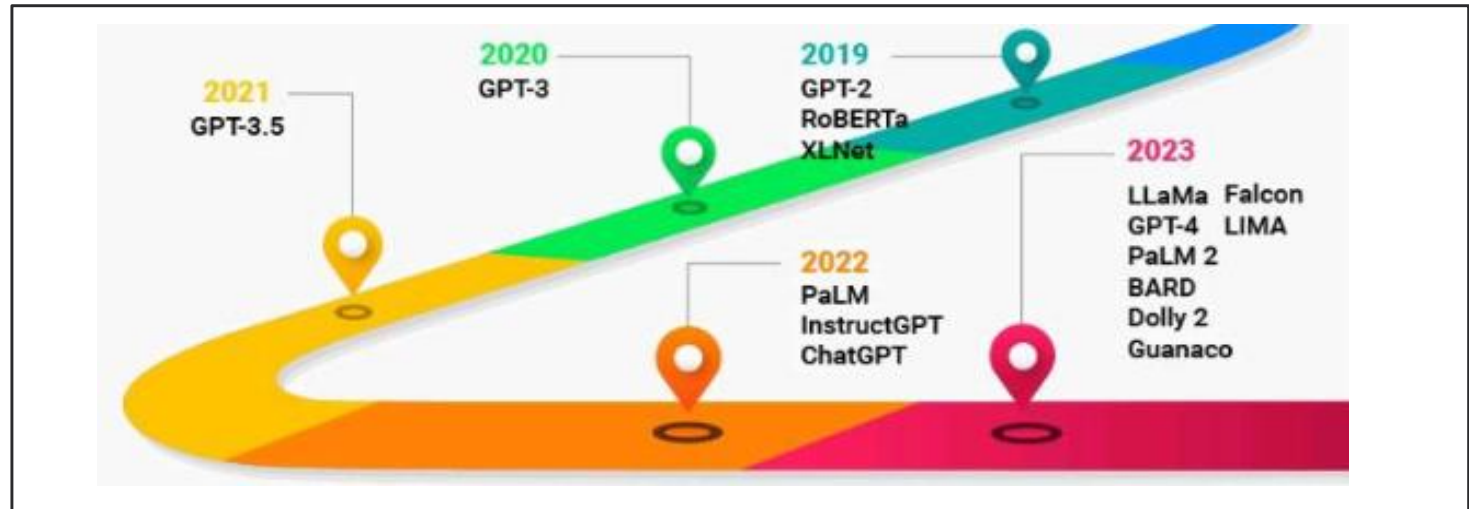
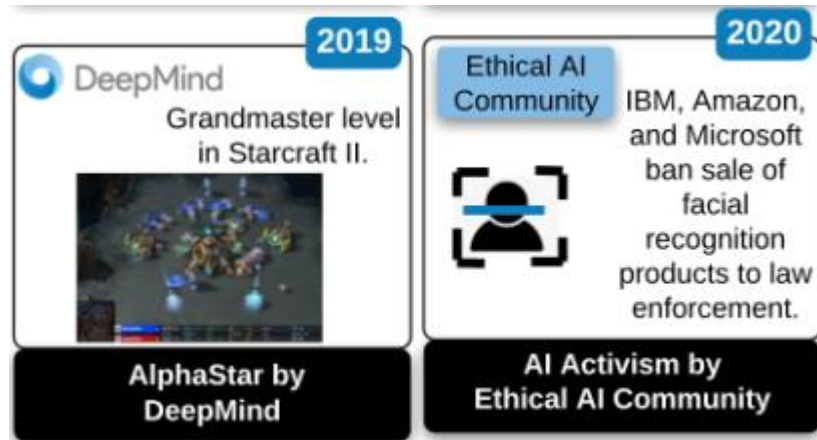


ImageNet by Fei-Fei Li

AI Evolution



Large Language Models



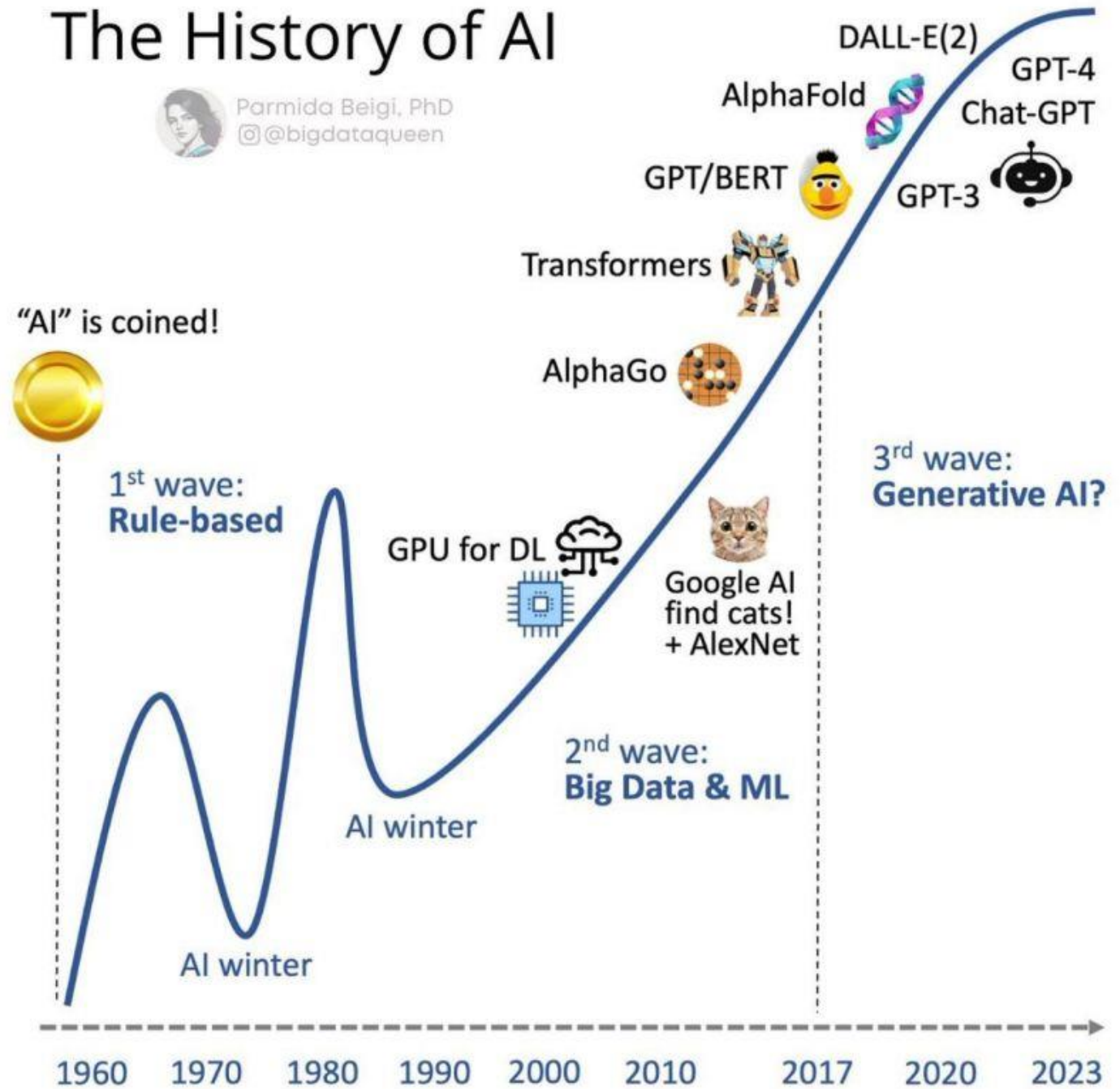
Winter of Machine Learning

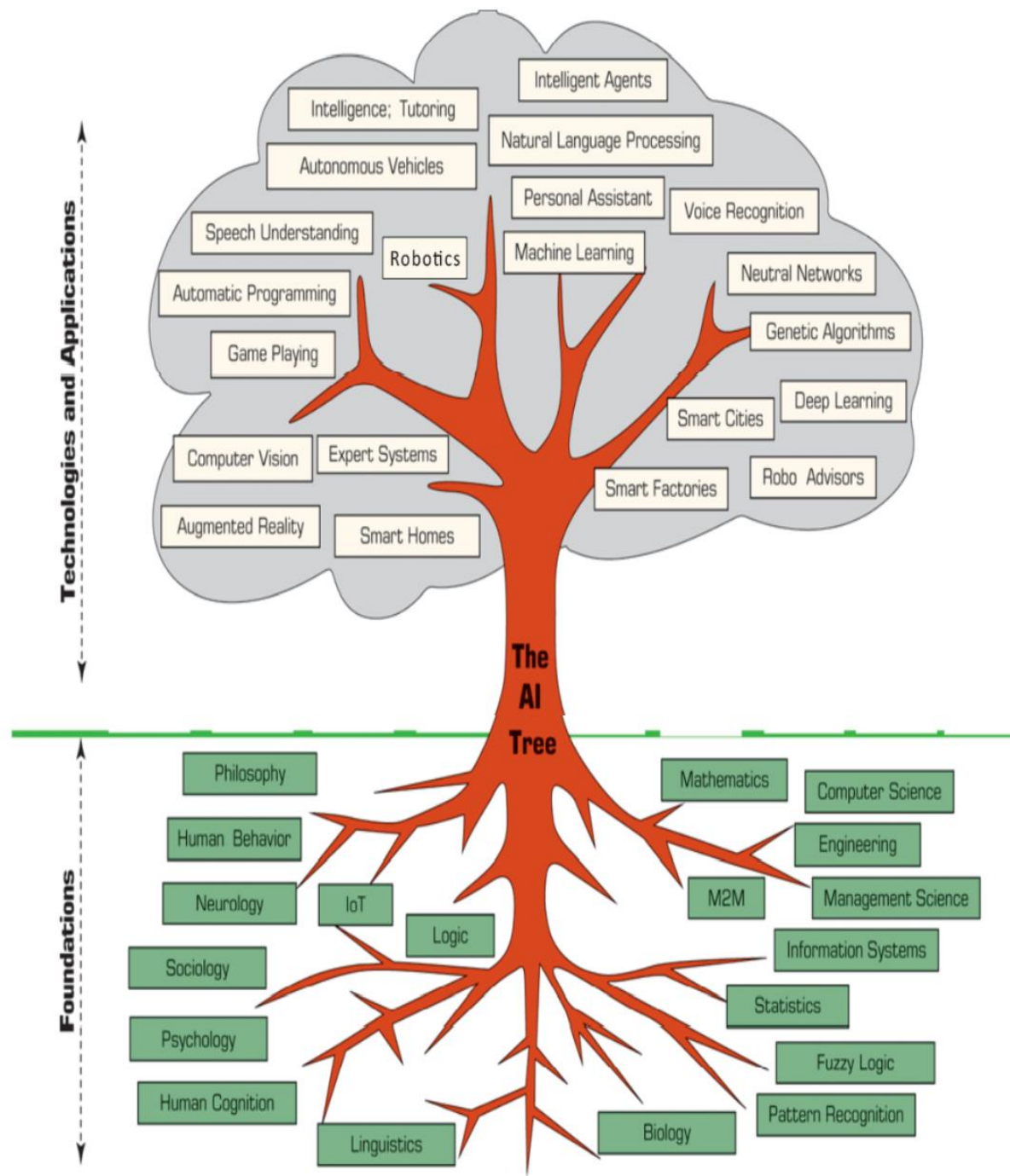
During the 1970s and 1980s, AI experienced what is known as the AI winter. Progress was slower than anticipated, and public interest waned. Funding for AI research decreased, leading to a period of reduced optimism and limited advancements.

The History of AI



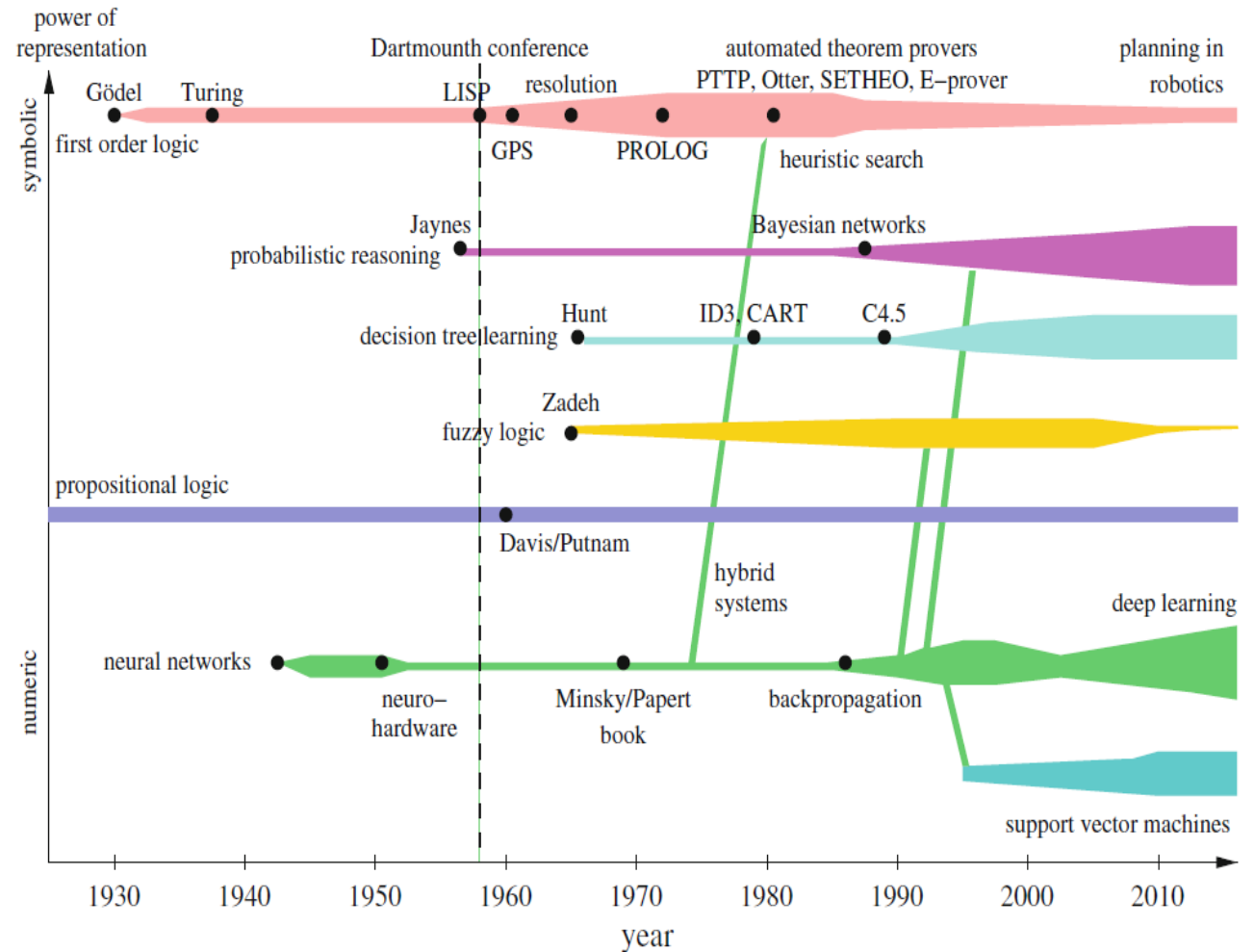
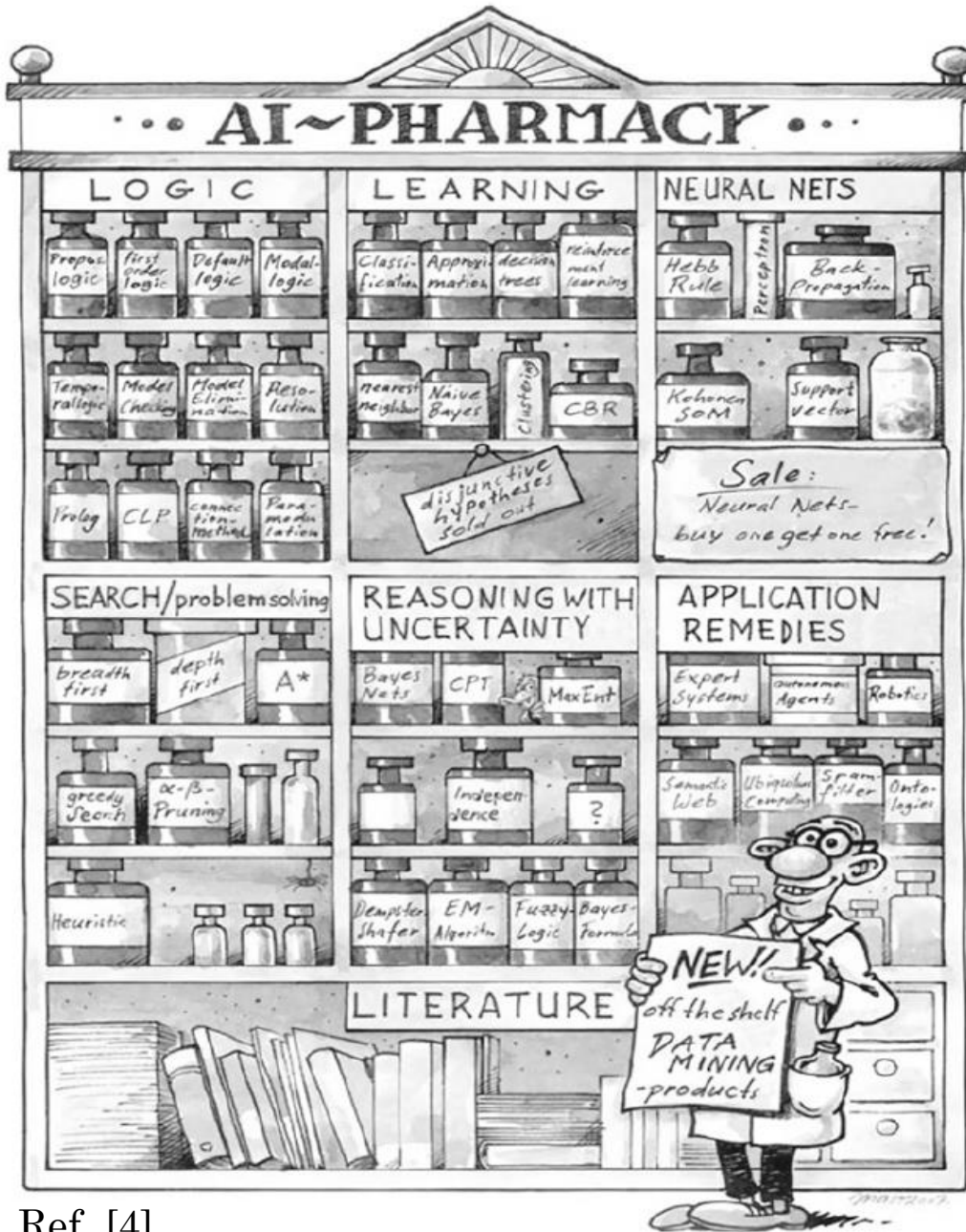
Parmida Beigi, PhD
@bigdataqueen





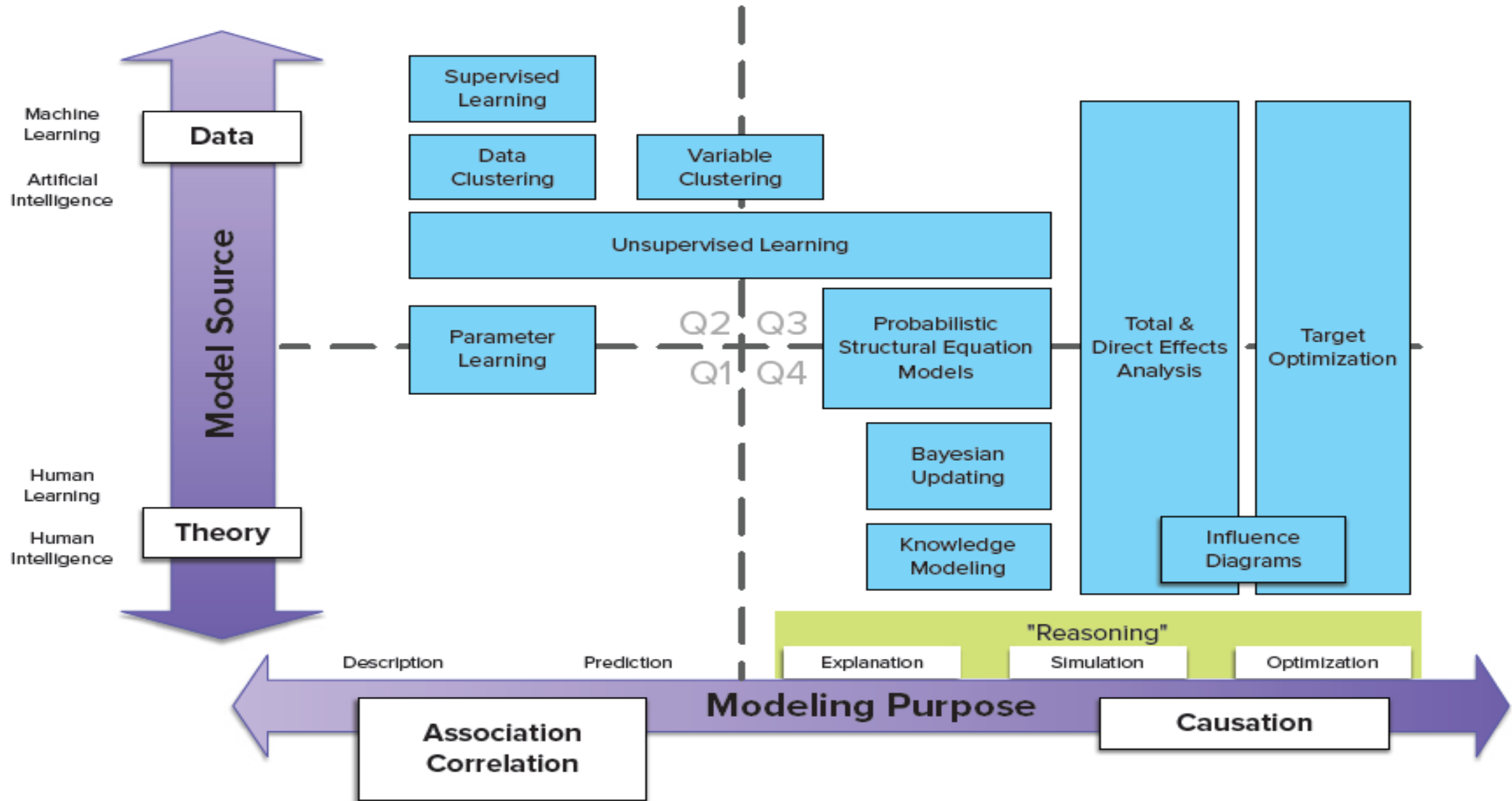
The AI Tree

AI Techniques



History of the various AI techniques. The width of the bars indicates prevalence of the techniques use.

AI Techniques: Modelling Domain



Recommended Reading

Wolfgang Ertel

Introduction to Artificial Intelligence

Second Edition

AI Classification



STRONG AI

AKA artificial general intelligence, an AI system with generalized human cognitive abilities. When presented with an unfamiliar task, it has enough intelligence to find a solution.



WEAK AI

AKA narrow AI, an AI system that is designed and trained for a particular task. Example: a virtual personal assistant, such as Apple's Siri.



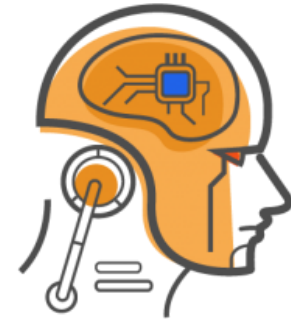
Narrow AI

Dedicated to assist with or take over specific tasks.



General AI

Takes knowledge from one domain, transfers to other domain.



Super AI

Machines that are an order of magnitude smarter than humans.

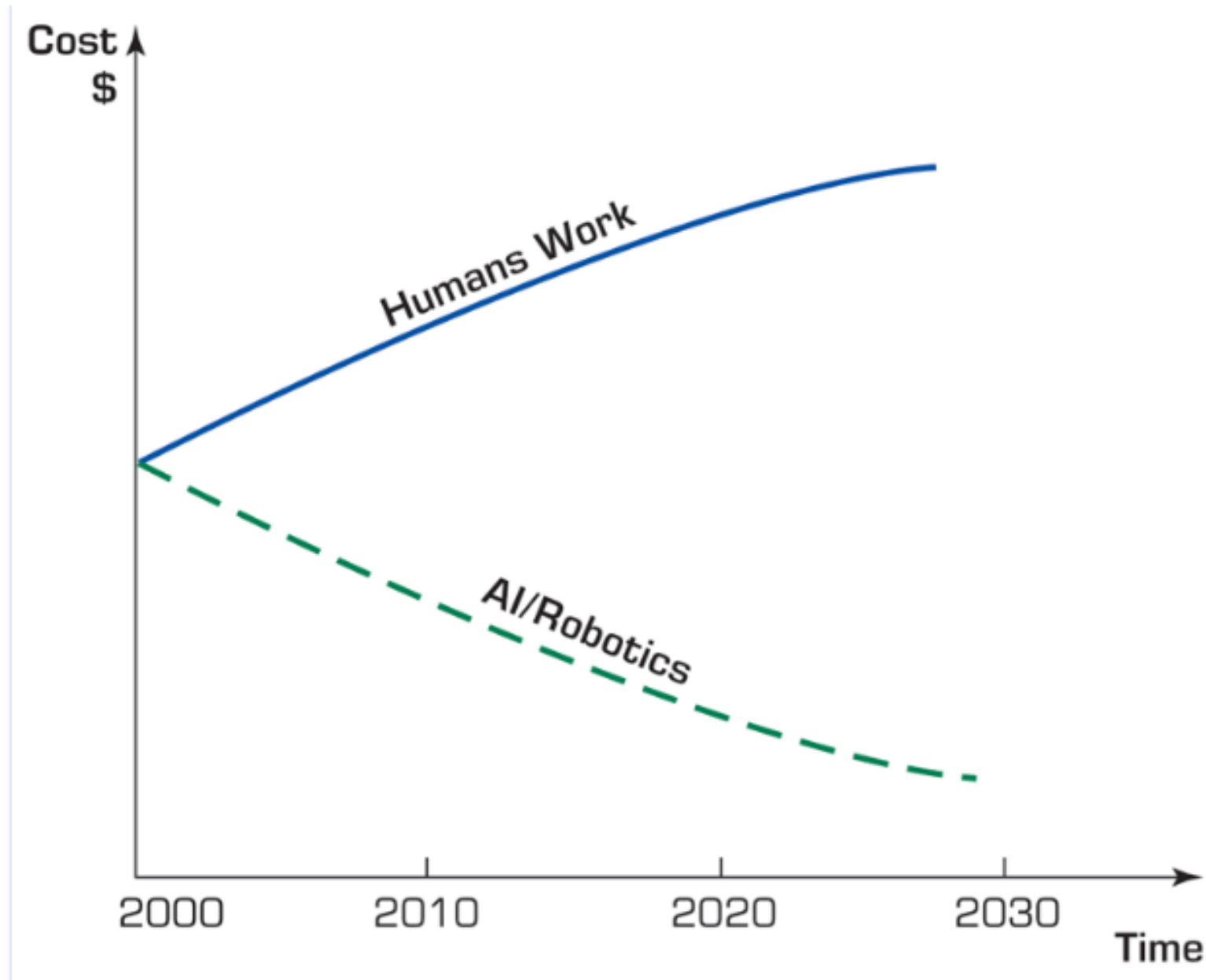
AI Claim

“You insist that there is something a machine cannot do. If you will tell me *precisely* what it is that a machine cannot do, then I can always make a machine which will do just that”

John von Neumann, 1948

- **Speed of execution** – While one doctor can make a diagnosis in ~10 minutes, AI system can make a million for the same time.
- **Less Biased** – They do not involve Biased opinions on decision making process.
- **Operational Ability** – They do not expect halt in their work due to saturation.
- **Accuracy** – Preciseness of the output obviously increases.

Cost Comparison AI vs. Human



11 Times AI Beat Humans at Games, Art, Law and Everything in Between

AI is simply better than humans at lots of tasks, from the obvious of data analysis to the unexpected of music making.

- The Deep Blue computer is a better chess-player since 1997
- Question and Answer Computer Watson took down Jeopardy Champion, 2011
- AI is faster and smarter at visual recognition
- An AI music program writes better pop songs: Flow Machines
- AI system is more performant at strategy games too, 2016
- AI creates and critics original art
- AI just smashed the record for solving the Rubik's Cube: 0.38 secs

AI vs Humans: Where AI Outperformed Us

From games to medicine, here's where artificial intelligence leads the way.



Games

AlphaGo beat world Go champion (2016)

OpenAI Five beat Dota 2 pros (2019)

Libratus crushed poker players (2017)



Knowledge & reasoning

Watson won Jeopardy (2011)

GPT-4 passes bar and SAT exams

Project Debater argues with humans on complex topics



Vision & health

AI radiologists outperform in diagnosing pneumonia, tumors

Self-driving cars detect obstacles faster than humans

AlphaFold solved protein folding challenge



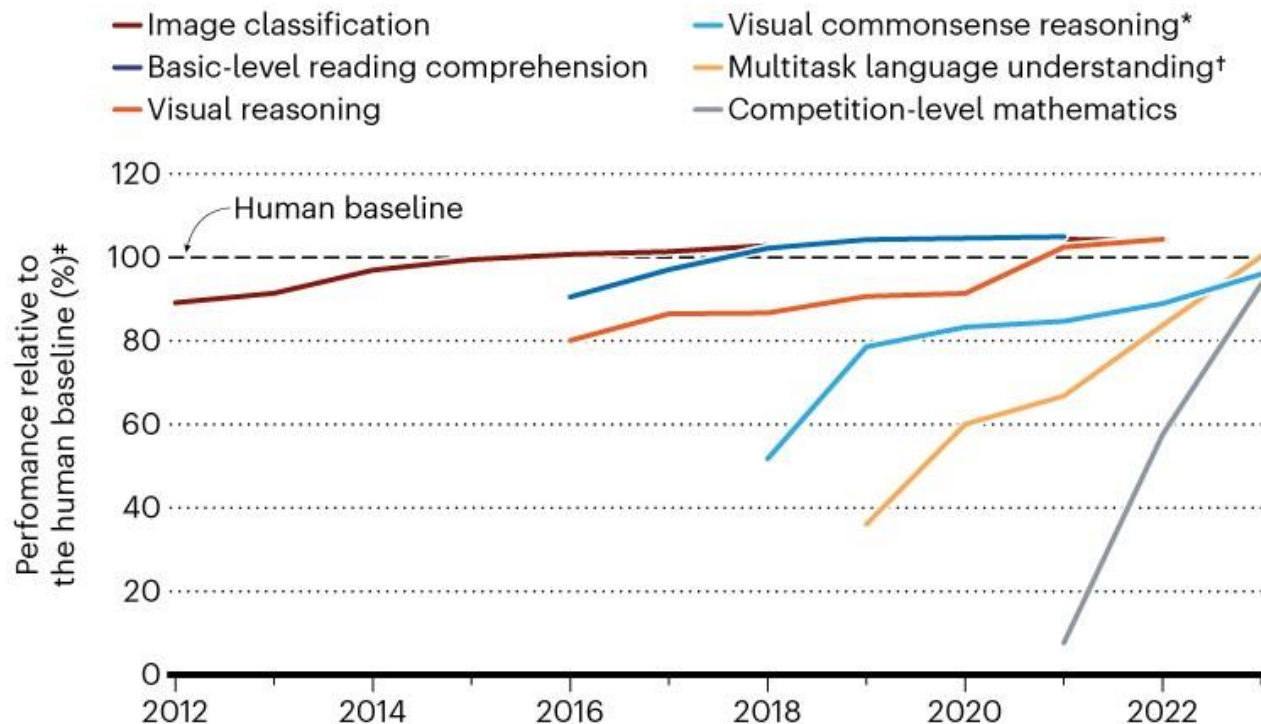
Creativity

Flow Machines composes pop songs

DALL-E / Midjourney generate realistic art

SPEEDY ADVANCES

In the past several years, some AI systems have surpassed human performance on certain benchmark tests, and others have made rapid progress.



*Requires an AI system to answer questions about an image and provide a rationale for why its answers are true.

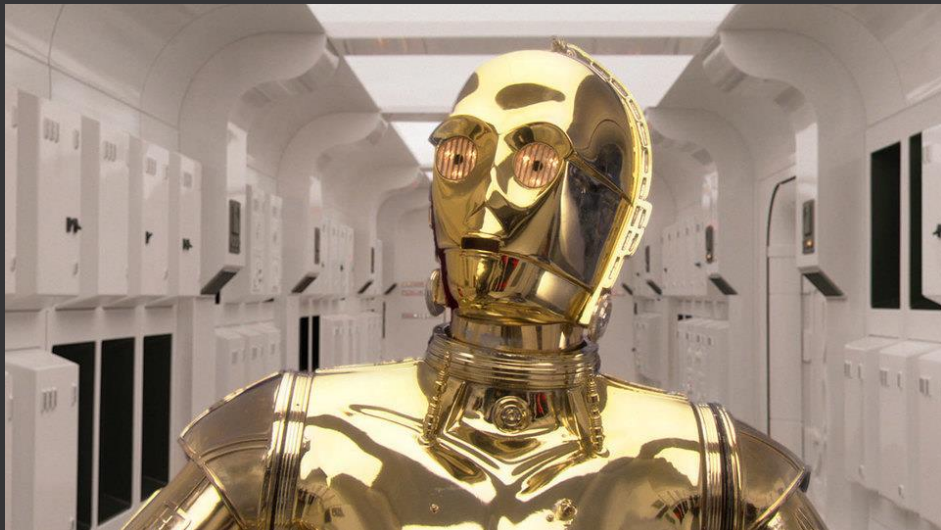
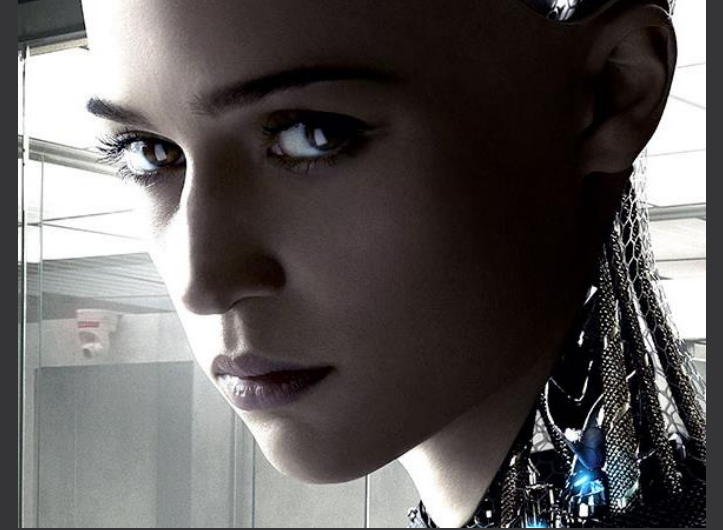
†Tests an AI model's knowledge and problem-solving ability with regard to 57 subjects, including broader topics such as mathematics and history, and narrower areas such as law and ethics.

*Data indicate the best performance of an AI model that year.

©nature

Ref. [6]

AI Isn't Exactly What we Thought it Was Going to be



Or Is It Going to be????

THE INTELLIGENCE EXPLOSION

WHEN AI BEATS HUMANS
AT EVERYTHING

JAMES BARRAT
Author of Our Final Invention

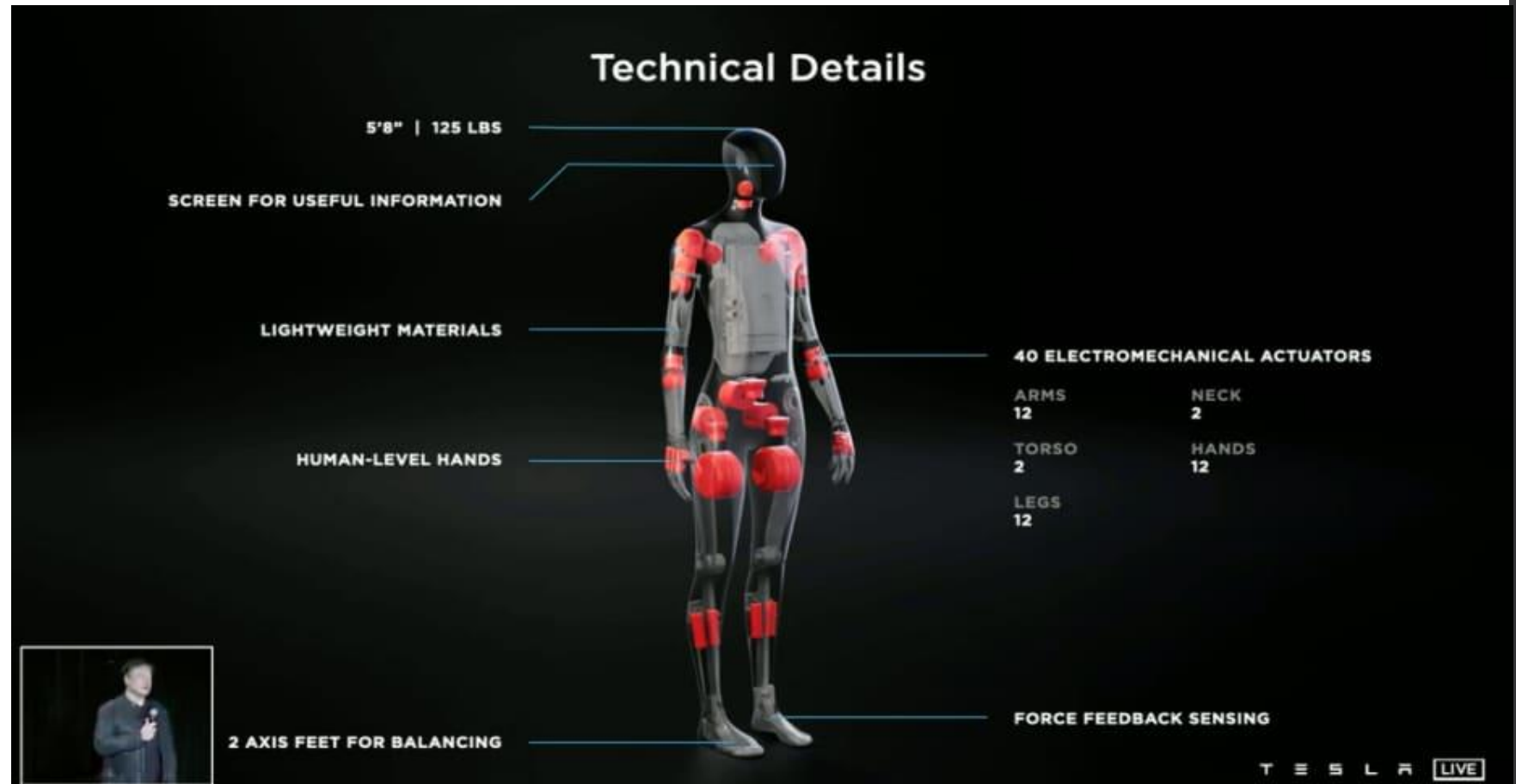


Tesla Bot

TECH

Elon Musk says Tesla will build a humanoid robot prototype by next year

PUBLISHED THU, AUG 19 2021-10:56 PM EDT | UPDATED FRI, AUG 20 2021-11:13 AM EDT



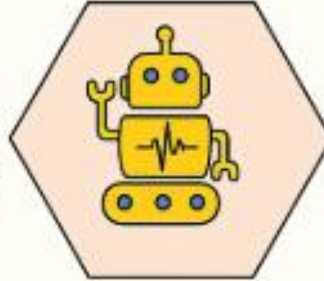
**Machine
Learning**



**Natural
Language
Processing**



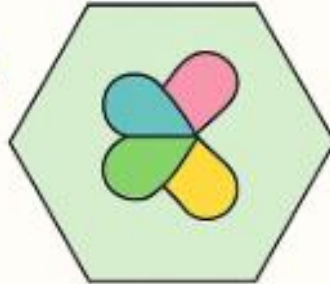
**Automation
& Robotics**



Deep Learning



**Generative
AI**



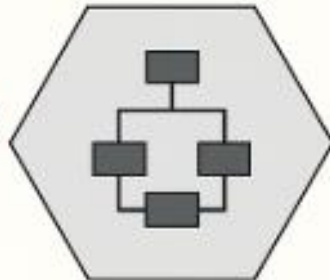
**Reinforcement
Learning**



**Computer
Vision**



**Expert
Systems**



AI Ethics



Top AI Techniques

AI vs. ML vs. DL

Artificial Intelligence



Making intelligent machines and programs

Machine Learning



Enabling model to learn without being explicitly programmed

Deep Learning



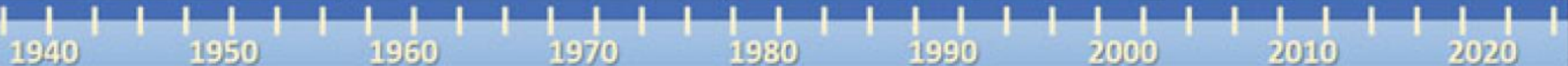
Special sub-discipline of ML based on neural based learning mimicking human cognitive abilities

Learning techniques for NN

Backpropagation

Genetic Algorithm

Swarm Optimization



Reference

- [1] <https://www.kidscodecs.com/what-is-artificial-intelligence/>
- [2] <https://techswizard.com/gadget/artificial-intelligence-vs-human-intelligence/>
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